

1. tumor_types

This table stores the different tumor types.

Column	Type	Description
id	INTEGER	Primary Key (Auto Increment)
type_name	TEXT	Name of tumor type (Unique)

2. tumor_grades

This table stores tumor grades.

Column	Type	Description
id	INTEGER	Primary Key (Auto Increment)
grade	TEXT	Grade of tumor (Unique)

3. locations

Stores different image location labels.

Column	Type	Description
id	INTEGER	Primary Key
location_name	TEXT	Location name (Unique)

4. images

This is the main table storing image metadata.

Column	Type	Description
id	INTEGER	Primary Key
file_name	TEXT	Image file name
width	INTEGER	Image width
height	INTEGER	Image height
x	INTEGER	X coordinate of annotation point
y	INTEGER	Y coordinate of annotation point
description	TEXT	Image description
type_id	INTEGER	Foreign Key → tumor_types(id)
grade_id	INTEGER	Foreign Key → tumor_grades(id)

5. image_locations

Junction table (Many-to-Many relationship between images and locations).

Column	Type	Description
image_id	INTEGER	Foreign Key → images(id)
location_id	INTEGER	Foreign Key → locations(id)

Primary Key: (image_id, location_id)

Query:

```
Query.py > ...
1  import sqlite3
2
3  conn = sqlite3.connect("brain_tumor_db.sqlite")
4  cursor = conn.cursor()
5
6  cursor.execute("SELECT name FROM sqlite_master WHERE type='table';")
7  print(cursor.fetchall())
```

OUTPUT DEBUG CONSOLE PROBLEMS ... Filter (e.g. text, !excludeText, t... Code ▾ ☰

[Running] python -u "c:\Users\ASUS\Desktop\My Project\Data Science\Final\phase 2\Query.py"

[('tumor_types',), ('sqlite_sequence',), ('tumor_grades',), ('images',), ('locations',), ('image_locations',)]

```
Query.py > ...
1  import sqlite3
2
3  conn = sqlite3.connect("brain_tumor_db.sqlite")
4  cursor = conn.cursor()
5
6  cursor.execute("PRAGMA table_info(images);")
7  print(cursor.fetchall())
```

OUTPUT DEBUG CONSOLE PROBLEMS 1 TERMINAL PORTS Filter (e.g. text, !excludeText, t... Code ▾ ☰ 🔗 ... | ⌂

[Running] python -u "c:\Users\ASUS\Desktop\My Project\Data Science\Final\phase 2\Query.py"

[(0, 'id', 'INTEGER', 0, None, 1), (1, 'file_name', 'TEXT', 1, None, 0), (2, 'width', 'INTEGER', 0, None, 0), (3, 'height', 'INTEGER', 0, None, 0), (4, 'x', 'INTEGER', 0, None, 0), (5, 'y', 'INTEGER', 0, None, 0), (6, 'description', 'TEXT', 0, None, 0), (7, 'type_id', 'INTEGER', 0, None, 0), (8, 'grade_id', 'INTEGER', 0, None, 0)]

```
Query.py > ...
1  import sqlite3
2
3  conn = sqlite3.connect("brain_tumor_db.sqlite")
4  cursor = conn.cursor()
5
6  cursor.execute("""
7  SELECT t.type_name, g.grade, COUNT(*) as count
8  FROM images i
9  JOIN tumor_types t ON i.type_id = t.id
10 JOIN tumor_grades g ON i.grade_id = g.id
11 GROUP BY t.type_name, g.grade;
12 """)
13 print(cursor.fetchall())
```

OUTPUT DEBUG CONSOLE PROBLEMS 1 TERMINAL PORTS Filter (e.g. text, !excludeText, t... Code ▾ ☰ 🔗 ... | ⌂ ✕

[Running] python -u "c:\Users\ASUS\Desktop\My Project\Data Science\Final\phase 2\Query.py"

[('Astrocytoma', 'T1', 396), ('Astrocytoma', 'T1C+', 441), ('Astrocytoma', 'T2', 281), ('Ependymoma', 'T1', 313), ('Ependymoma', 'T1C+', 374), ('Ependymoma', 'T2', 350), ('Glioma', 'T1', 522), ('Glioma', 'T1C+', 549), ('Glioma', 'T2', 405), ('Hemangiopericytoma', 'T1', 178), ('Hemangiopericytoma', 'T1C+', 307), ('Hemangiopericytoma', 'T2', 118), ('Meningioma', 'T1', 636), ('Meningioma', 'T1C+', 977), ('Meningioma', 'T2', 458), ('Neurocytoma', 'T1', 190), ('Neurocytoma', 'T1C+', 254), ('Neurocytoma', 'T2', 174), ('Normal', 'T1', 415), ('Normal', 'T1C+', 264), ('Normal', 'T2', 379), ('Oligodendroglioma', 'T1', 221), ('Oligodendroglioma', 'T1C+', 192), ('Oligodendroglioma', 'T2', 137), ('Other', 'T1', 412), ('Other', 'T1C+', 756), ('Other', 'T2', 365), ('Schwannoma', 'T1', 363), ('Schwannoma', 'T1C+', 564), ('Schwannoma', 'T2', 309)]